

Exploring disentangled latent states for World Models

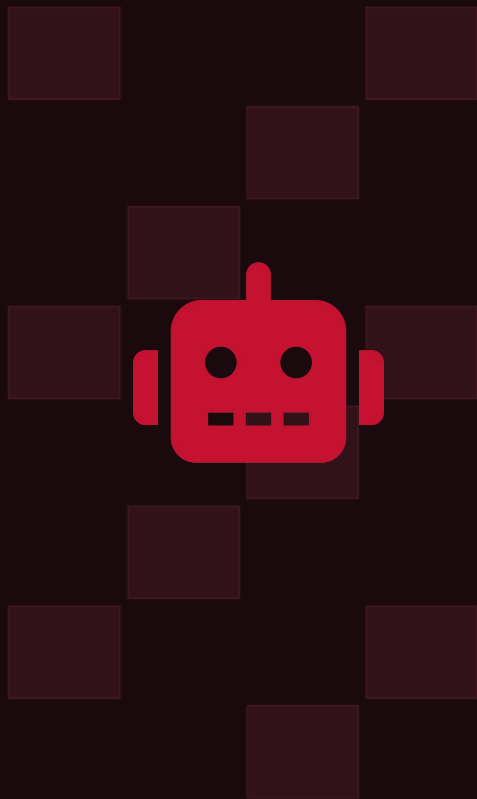
Group 4: Yiqi Wang · Nate Ludlow · Tony Wu · Teddy Lin

April 15, 2026

*Static scene structure doesn't change when the robot acts.
Why predict it over and over?*

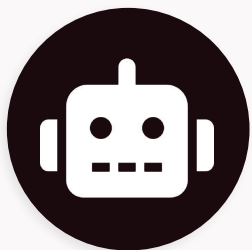
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Robotics Institute · 16-761



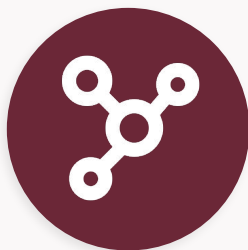
What is a World Model?

Learn to predict the consequences of actions and environment dynamics and use for planning



Observe

Raw pixels o_t + State x_t



Encode

Latent z_t



Predict

Next state z_{t+1}



Plan

MPC / MPPI



Why It Works

Models are trained on large offline datasets to predict what happens next given an action, learning the rules of the environment without needing to interact with it at test time.



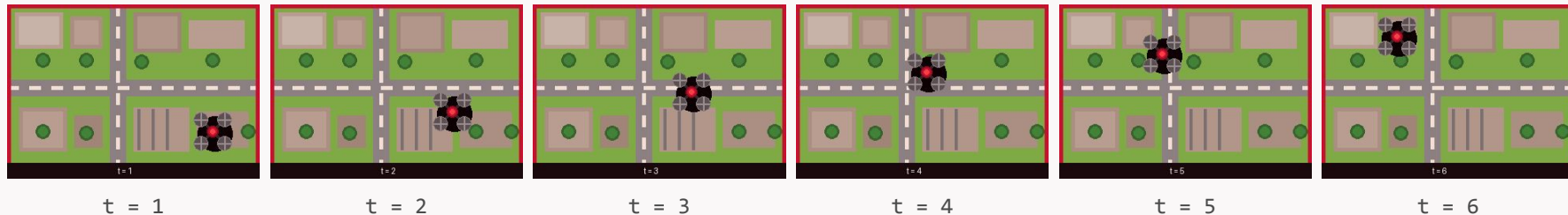
The Problem

Dense prediction: the world model model re-predicts *everything* at every step — even parts of the scene that never change.

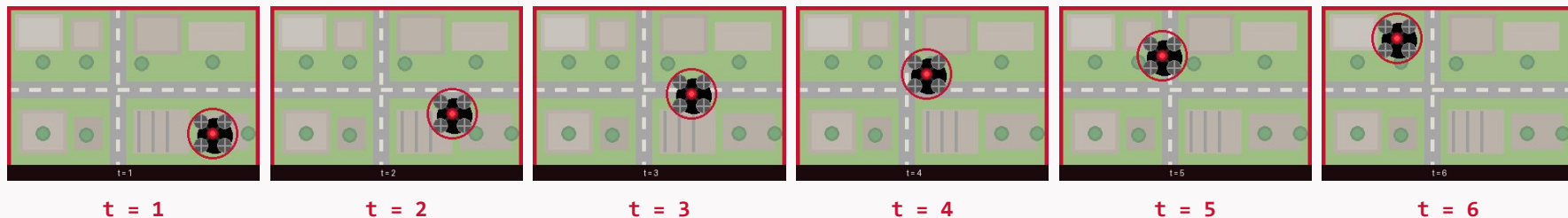
The Problem: Re-Predicting What Doesn't Change

Dense WM — entire scene re-predicted every step

But the whole background is static!



Our Disentangled WM — background encoded once with minimal changes, only drone movement predicted



Spend the most compute on predicting what changes and little compute on what doesn't.

The Fix: Split Static and Dynamic State

Much of what vision sees is static from frame to frame, often only a small part is truly dynamic.

BEFORE

Dense World Model

Full state z_t

 Background +  robot +  objects — all mixed



🔄 *re-predict every timestep*

Full state z_{t+1}

 Background *re-computed, even though nothing changed*



Complex prediction of more objects is challenging
→ slow, inaccurate planning

AFTER

Disentangled WM (Ours)



Static s_t

Constant frame observations

s_t static ↓



Dynamic e_t

Robot, objects
evolves each step

e_t dynamic ↓

$$e_{t+1} = f_{\theta}(s_t, e_t, a_t)$$

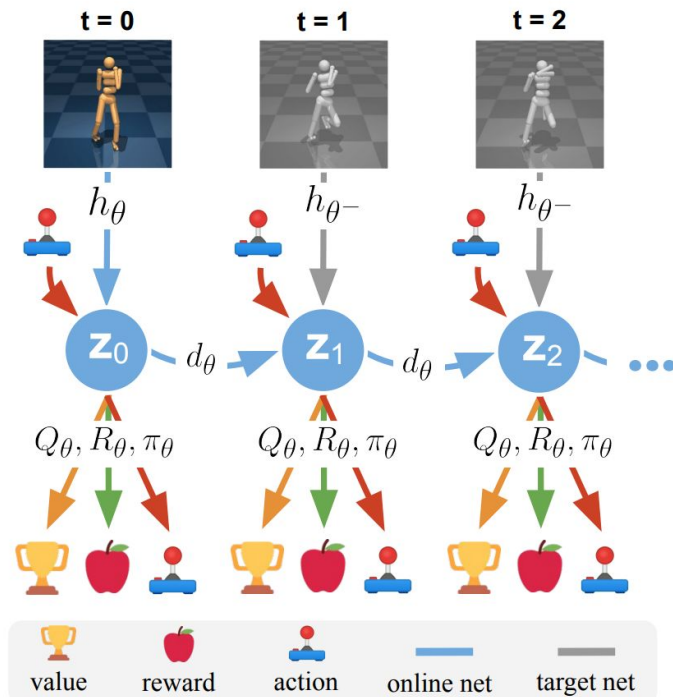
✓ Faster planning

✓ Less error

✓ Scalable

Related Work: TD-MPC

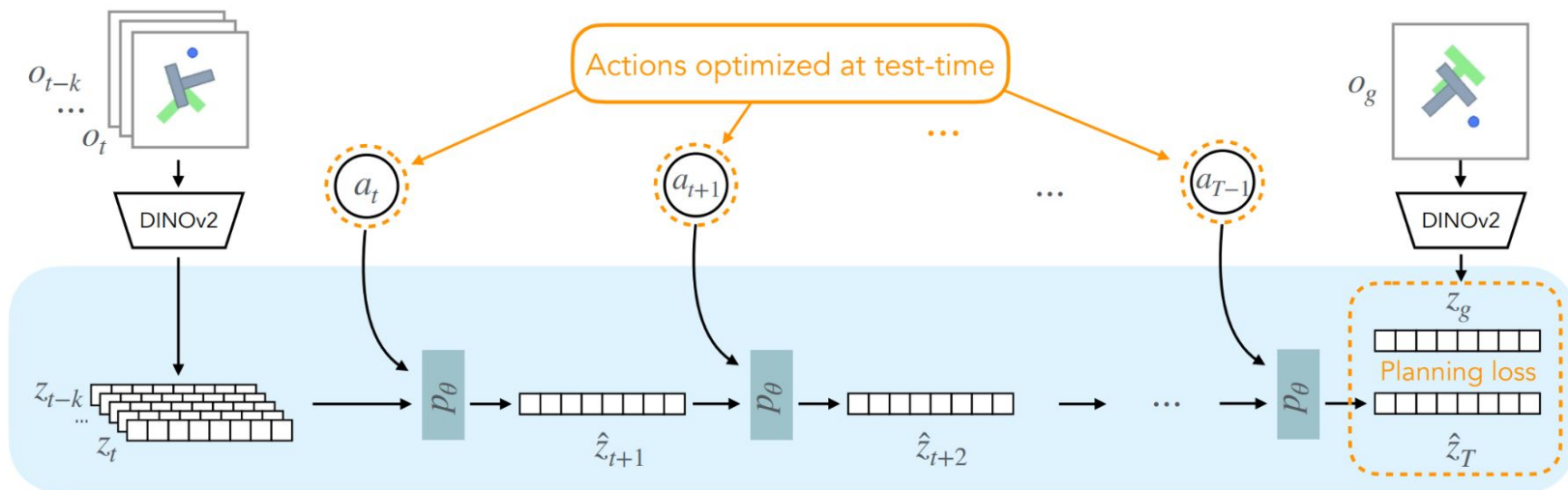
- Jointly learning latent dynamics, encoders and terminal value with temporal difference learning.



Representation: $\mathbf{z}_t = h_\theta(\mathbf{s}_t)$
Latent dynamics: $\mathbf{z}_{t+1} = d_\theta(\mathbf{z}_t, \mathbf{a}_t)$
Reward: $\hat{r}_t = R_\theta(\mathbf{z}_t, \mathbf{a}_t)$
Value: $\hat{q}_t = Q_\theta(\mathbf{z}_t, \mathbf{a}_t)$
Policy: $\hat{\mathbf{a}}_t \sim \pi_\theta(\mathbf{z}_t)$

Related Work: DINO-WM

- Predict next latent state with a history of past latent states
- Predict the full latent state



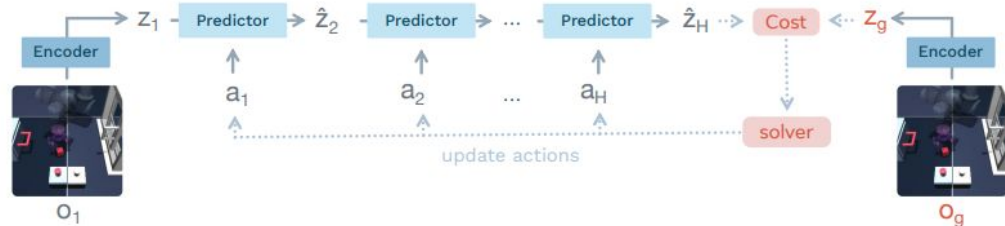
Baseline: LeWorldModel

Learns a latent world model with an visual encoder (ViT).

Predictor models dynamics in latent space by predicting the next frame embedding given latent embedding z_t and action a_t .

Objective is a sum of prediction and regularization loss

Performs latent-space planning by simulating trajectories and minimizing goal distance



Encoder: $z_t = \text{enc}_\theta(o_t)$

Predictor: $\hat{z}_{t+1} = \text{pred}_\phi(z_t, a_t)$

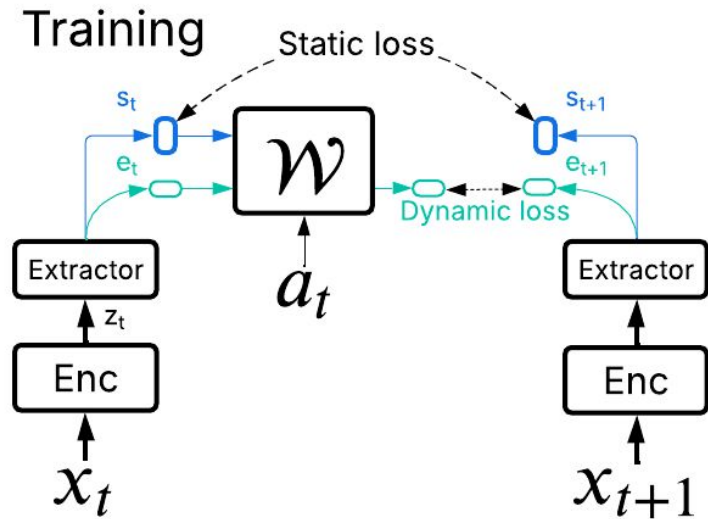
$\mathcal{L}_{\text{LeWM}} \triangleq \mathcal{L}_{\text{pred}} + \lambda \text{SIGReg}(\mathbf{Z})$

$\mathcal{L}_{\text{pred}} \triangleq \|\hat{z}_{t+1} - z_{t+1}\|_2^2$ $\text{SIGReg}(\mathbf{Z}) \triangleq \frac{1}{M} \sum_{m=1}^M T(h^{(m)})$

$\mathbf{a}_{1:H}^* = \arg \min_{\mathbf{a}_{1:H}} \mathcal{C}(\hat{z}_H)$

$\mathcal{C}(\hat{z}_H) = \|\hat{z}_H - z_g\|_2^2$, $z_g = \text{enc}_\theta(o_g)$

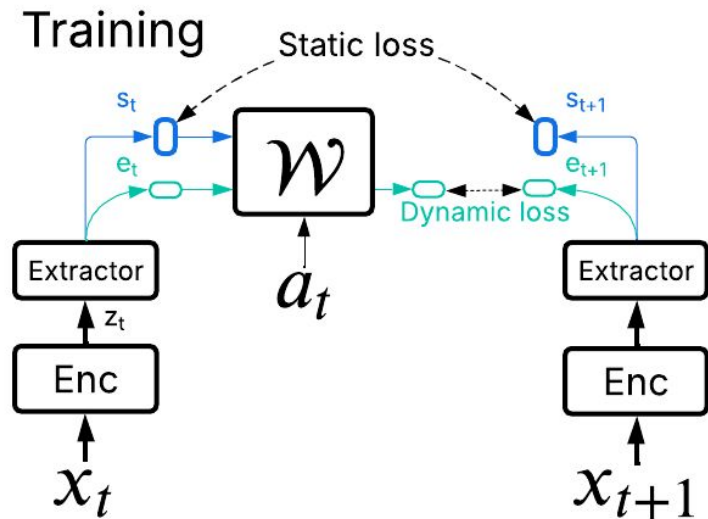
Our Approach



We inherit LeWorldModel's loss for learning the dynamics states: e_t .

- State extractor: a cross-attention layer.

Our Approach

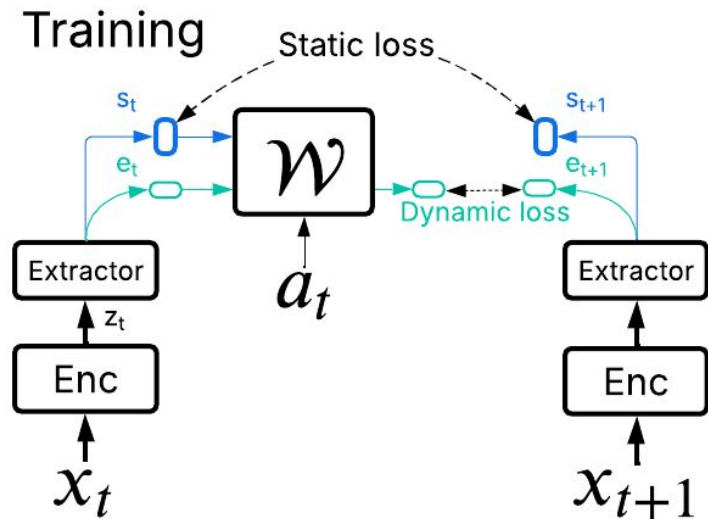


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Static loss = $w_{ij} \cdot ||e_i, e_j||_2^2$ where w_{ij} is $1/\text{abs}(i-j)$.

Our Approach



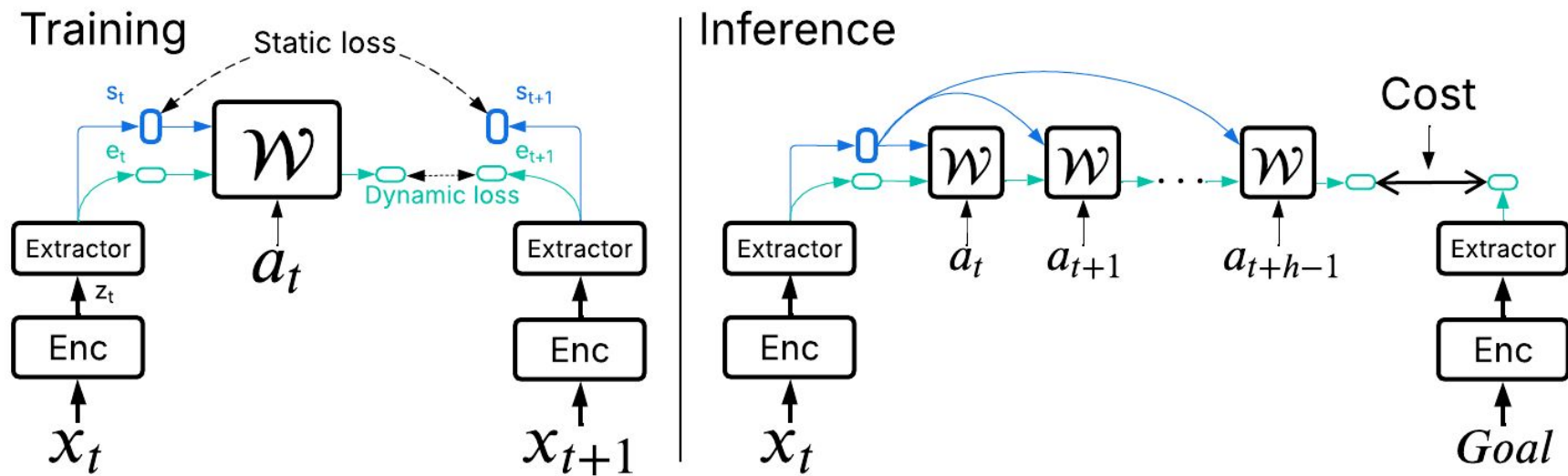
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Motivation: allowing the static state to update slowly from $i \rightarrow j$. Avoiding degenerated states.

Our Approach



Result — Task: TwoRoom

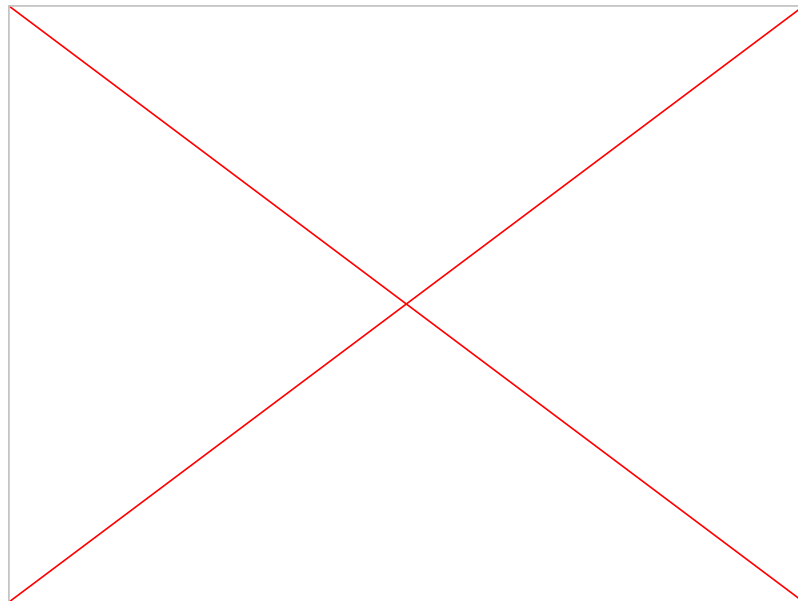
Task: tworoom

Dataset:

- 10000 episodes.
- 920k transitions.

Compute:

- Single NVIDIA RTX A6000, 2-3 days.



Result — Task: TwoRoom

Task: tworoom

Baselines:

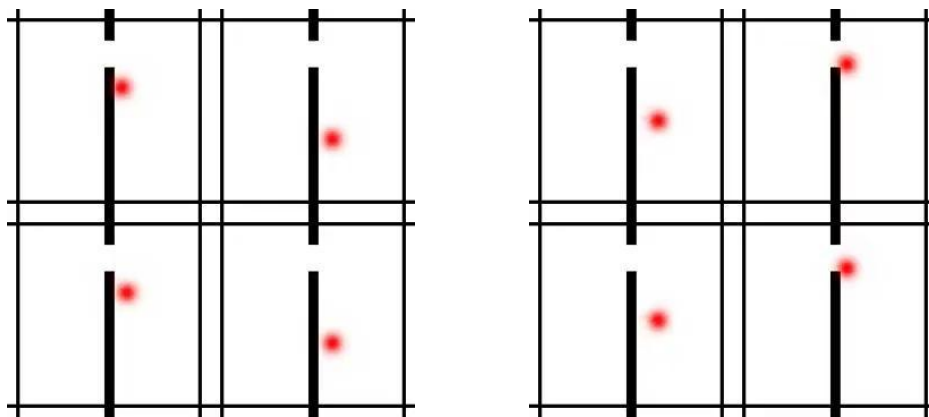
- LeWM-scratch: original LeWorldModel learns from scratch.
- LeWM-Dino: LeWorldModel with a frozen DINO encoder.
- LeWM-Dino-disentangled: LeWorldModel with a frozen DINO encoder, and state extractors for static and dynamic states.

Result — Task: TwoRoom

Task: tworoom

Baseline		Ours
LeWM-scratch	LeWM-Dino	LeWM-Dino-disentangled
78%	42%	46%

Typical successes

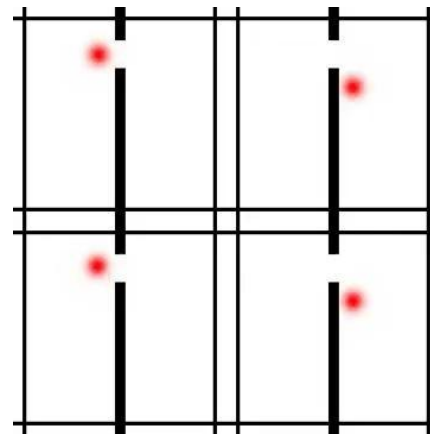
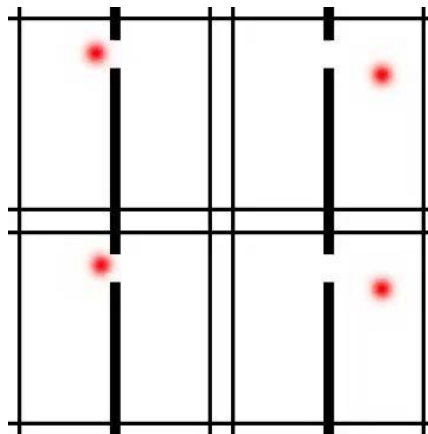


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Typical failures



Result — Task: TwoRoom

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Findings:

- States learns from scratch performs the best. ← Compute intensive.
- When given a frozen encoder (Dino):
 - Our proposed approach has moderate improvement.
 - The static state makes planning less vulnerable to compounding errors.

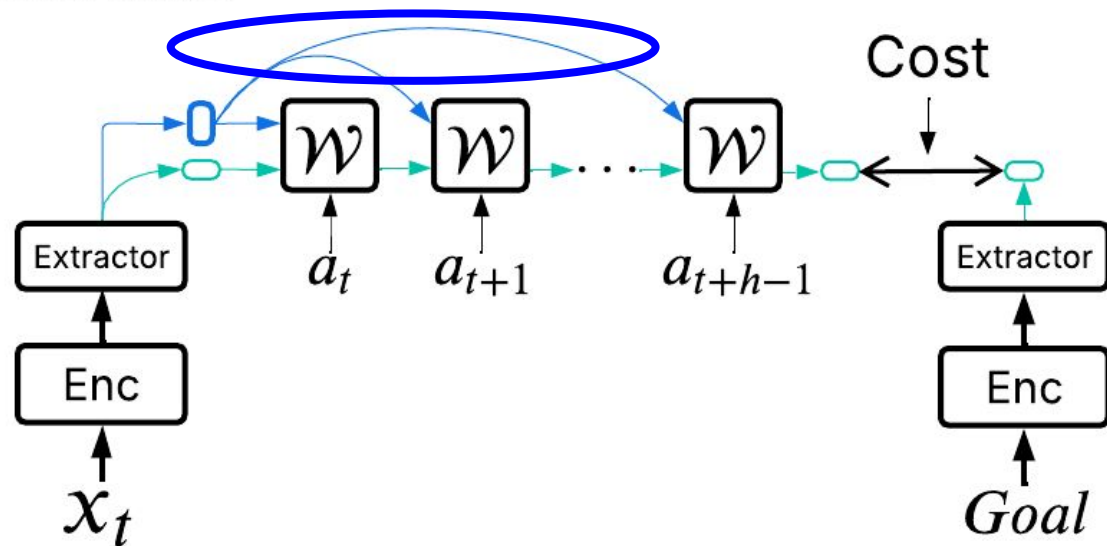
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Baseline	
LeWM-scratch	LeW
78%	4

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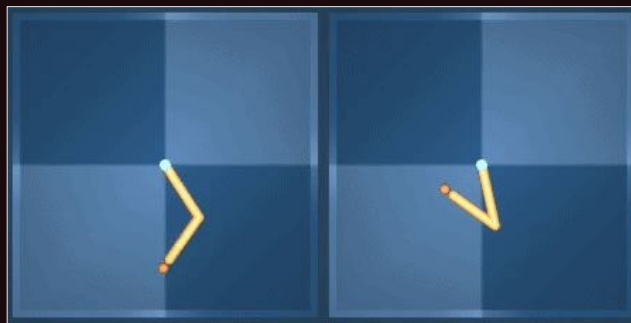
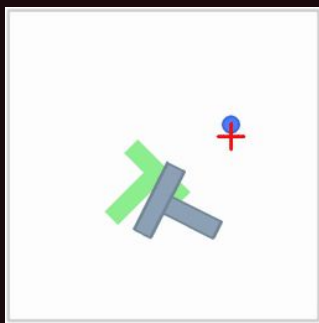
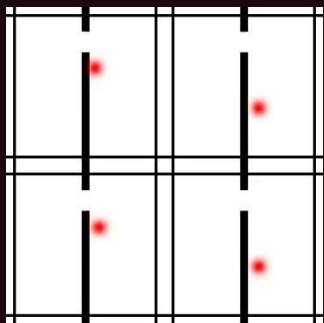
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This suggests that off-the-shelf pretrained representation lacks information sufficient for World Modeling, without further finetuning.

Next Steps

01 Verify our idea by learning dynamic/static state representations **from scratch**

02 Extend evaluations to more challenging domains



Questions?

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